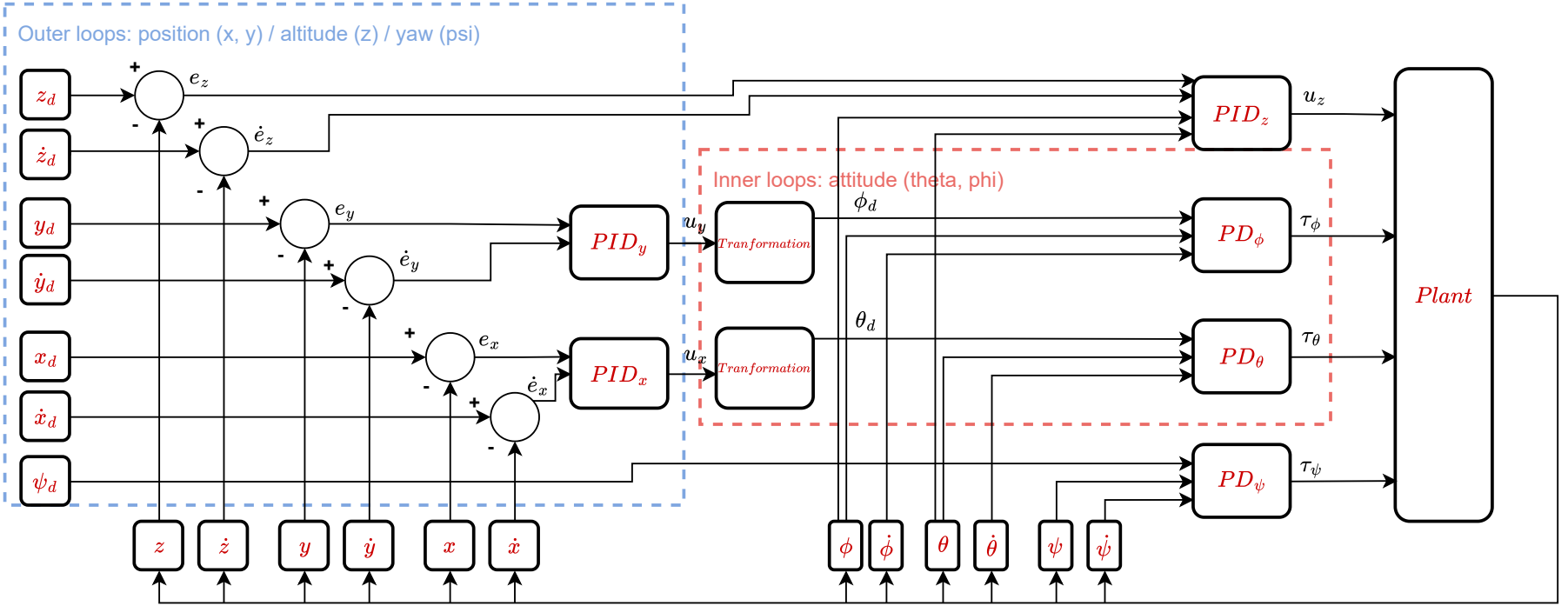

Outer loops: position (x, y) / altitude (z) / yaw (psi)
Inner loops: attitude (theta, phi)
z_d
$\dot{z}_d$
y_d
$\dot{y}_d$
x_d
$\dot{x}_d$
ψ_d
e_z
$\dot{e}_z$
e_y
$\dot{e}_y$
e_x
$\dot{e}_x$
PID_y
PID_x
u_y
u_x
Tranformation
Tranformation
ϕ_d
θ_d
PID_z
PD_ϕ
PD_θ
PD_ψ
u_z
τ_ϕ
τ_θ
τ_ψ
Plant
z
$\dot{z}$
y
$\dot{y}$
x
$\dot{x}$
ϕ
$\dot{\phi}$
θ
$\dot{\theta}$
ψ
$\dot{\psi}$